

SUHAAS CHOUDARY MALLAVARAPU

Software Engineering Intern, Computer Science Undergraduate

Narasaraopet, Ravipadu, Andhra Pradesh, India | +91-9290770994 | suhaaschoudarymallavarapu10@gmail.com
linkedin.com/in/suhaas-choudary-mallavarapu | github.com/suhaas2006

SUMMARY

Computer Science undergraduate seeking a Software Engineering Internship at Google. Proficient in Python and C with hands-on experience in data structures and algorithms, object-oriented programming, and scalable software development. Built 2 end-to-end software systems delivering 35% performance improvements and logarithmic query optimization across 1,000+ records. Strong problem solver and effective communicator with a collaborative work ethic, attention to detail, and a passion for writing efficient, clean, and maintainable code.

EDUCATION

B.Tech in Computer Science and Engineering

Aug 2024 - Present

SRM University AP, Andhra Pradesh | CGPA: 7.4

Coursework: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, System Design

Intermediate, Mathematics Physics and Chemistry (Class XII)

Apr 2022 - Mar 2024

Bhashyam Junior College, Andhra Pradesh | Score: 86%

ICSE (Class X)

Mar 2022

Loyola Public School | Score: 90%

SKILLS

Programming Languages: Python, C

Libraries and Tools: OpenCV, NumPy, PyMongo, Git, GitHub, VS Code, MongoDB Atlas, Command Line Interface (CLI)

Databases: MongoDB (NoSQL), aggregation pipelines, indexing, query optimization, CRUD operations

Technical Concepts: Data Structures and Algorithms, Object-Oriented Programming (OOP), System Design, Computer Vision, Scalable Pipeline Architecture, Performance Optimization, Modular Software Development, Backend Development, Software Development Life Cycle (SDLC), API Design, Debugging, Unit Testing, Version Control, Agile Development, Technical Documentation

Soft Skills: Problem Solving, Analytical Thinking, Effective Communication, Team Collaboration, Attention to Detail, Adaptability, Self-Motivated, Time Management, Leadership, Quick Learner

PROJECTS

Scalable 3D Reconstruction Pipeline | Python, OpenCV, NumPy

Jan 2024 - May 2024

- Built an end-to-end computer vision pipeline consisting of 4 decoupled modules including preprocessing, feature extraction using SIFT and ORB algorithms, descriptor matching, and 3D reconstruction, processing 300 or more images with consistent structural accuracy.
- Reduced per-image processing overhead by 35% by implementing batched image loading and lazy evaluation, enabling the system to scale to datasets twice as large without requiring any architectural changes.
- Improved 3D model accuracy by 40% by applying geometric verification to eliminate false feature correspondences, reducing structural noise in the final point cloud output across all test cases.
- Designed 3 integration-compatible interfaces targeting medical imaging and spatial analysis use cases, demonstrating system design tradeoff analysis across computational cost, memory usage, and output precision.

Modular Student Record Management System | Python, C

Aug 2024 - Nov 2024

- Developed a full-stack backend system with complete CRUD functionality built on a 3-layer modular architecture that separates data access logic, business logic, and interface layers, following software engineering best practices throughout the software development life cycle.
- Improved data retrieval performance from linear to logarithmic time complexity by implementing hash-based indexing and sorted data structures, cutting average lookup time by 60% across 1,000 or more student records.
- Implemented atomic write operations with schema-consistent serialization logic, achieving zero data loss across 100 or more simulated session cycles and validating data integrity under concurrent update conditions.
- Reduced future feature integration effort by 50% by applying single-responsibility design and separation of concerns, enabling extensions such as role-based access control and reporting analytics with no regression to core data operations.

CERTIFICATIONS AND ACTIVITIES

- Earned MongoDB Associate Developer Certification in 2024, validating proficiency in NoSQL data modeling, aggregation pipeline design, indexing strategies, CRUD operations, and query performance optimization across 5 core assessment domains.
- Founded and managed a science and technology YouTube channel, producing 50 or more videos on space exploration and emerging technologies, demonstrating technical communication skills and a commitment to self-directed learning.